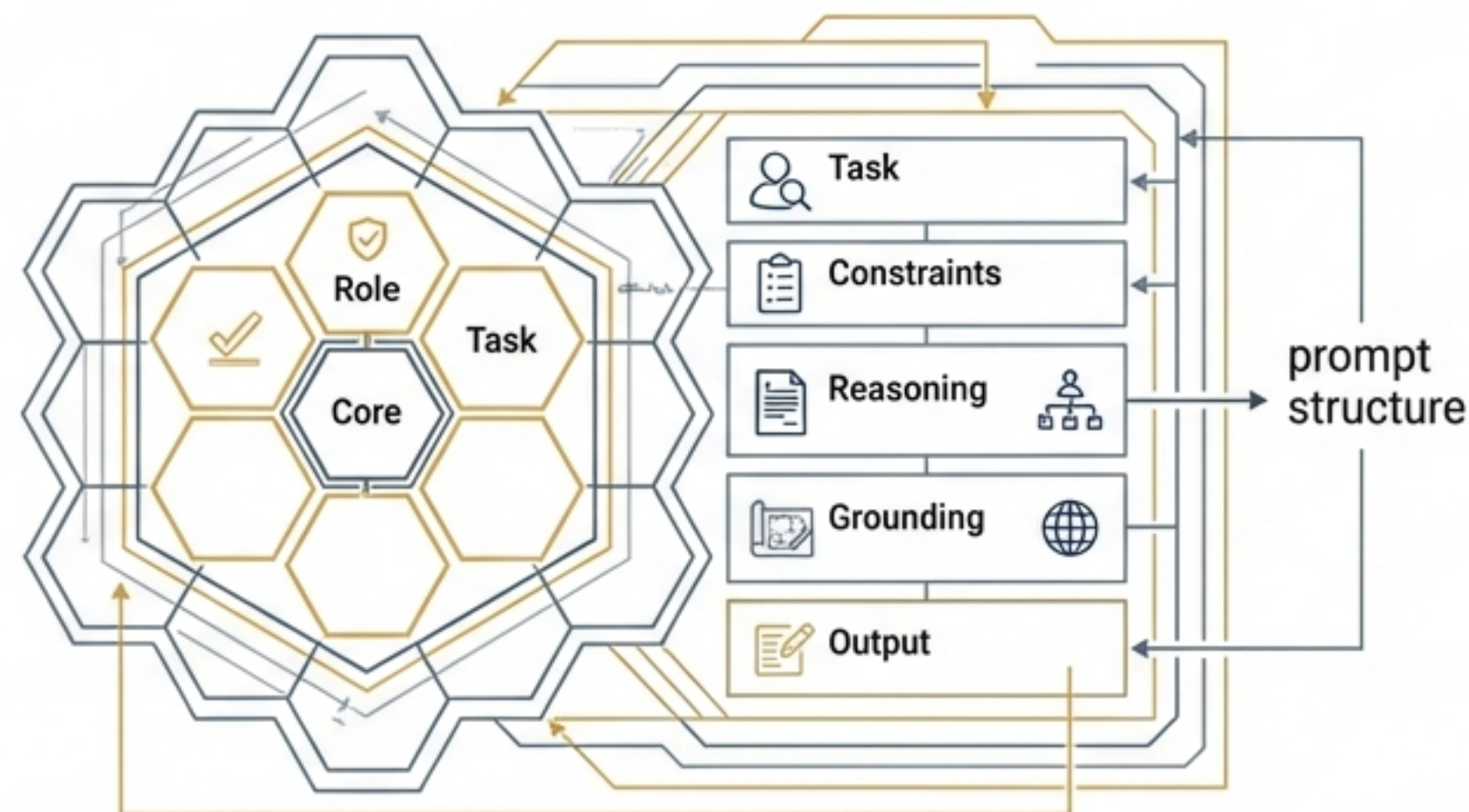




Fact-Check Your AI: A Framework for Verifiable Answers

From hallucination to hard evidence using the
Modular Prompt Construction (V2) framework



Reliability is the ceiling on AI utility.

AI models are probabilistic, not deterministic. Without strict architectural guidance, they prioritize plausibility over truth.

The Core Benefits



Enhances Credibility: Builds user trust by making tcy easy to confirm.



Promotes Accountability: Holds the AI accountable to primary references.



Facilitates Fast Verification: Allows users to quickly validate critical data points.

Who Needs This?



Journalism & Media



Finance & Business

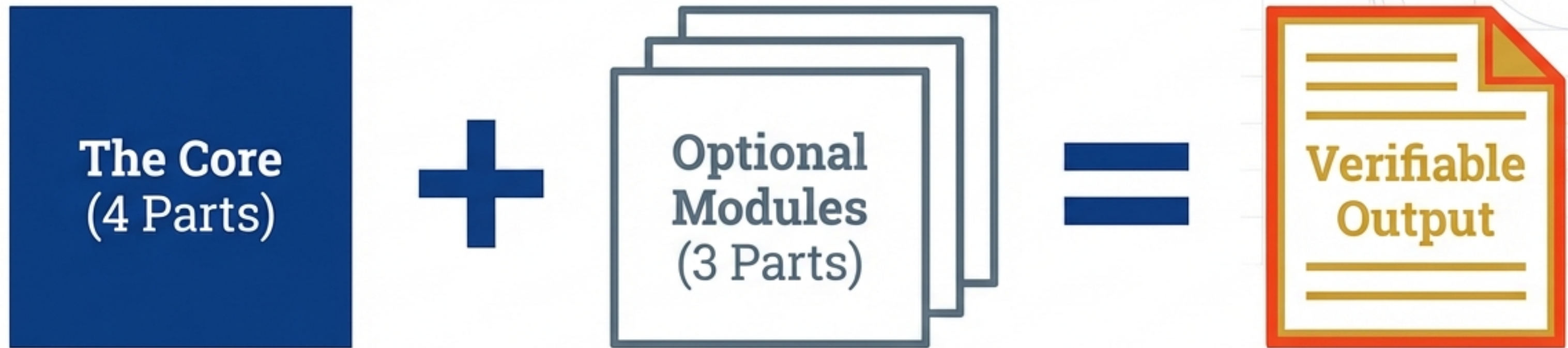


Legal & Healthcare



Research

The V2 Framework: Engineering reliability.



Key Concept: The Core is mandatory. Modules are used only when complexity or risk increases.



Core Component 1: Context

Theory

Definition

Foundational background the model needs for accuracy.

What to include

- Intended audience
- Key facts
- Scope boundaries

Specificity Levels

Low for simple queries; High for nuanced, domain-specific tasks.

Application

The Prompt Input

Context: I will provide a claim to verify. Assume the audience is a typical knowledge worker who needs a clear, evidence-based conclusion.

Target Audience



Core Component 2: Task (Goal)

Theory

Definition

The action the AI must perform, stated with active verbs.

Requirements

Clear deliverables and completion criteria (what "done" looks like).

Specificity

High specificity is required for multi-step work needing clear sequencing.

Application

The Prompt Input

Task (Goal): Determine whether the claim *Tunc* is True, False, or Unsubstantiated. Summarize the strongest supporting or refuting evidence and cite sources.

Discrete Deliverables



Core Component 3: Output Format

Theory

Definition

The structure the answer should follow.

Why it matters

Enforces readability and allows for easier human scanning.

Types

Bullets, tables, or strict schemas (JSON/Markdown).

Application

The Prompt Input

Control Mechanism

Output Format: Use exactly these headings:

- 1) Verdict: True / False / Unsubstantiated.
- 2) Evidence Summary: (≤ 70 words).
- 3) Sources: (APA-style list).



Core Component 4: Constraints & Tone

Theory

Inter Tight

Definition

Rules the output must follow, including limitations and safety.

Key Elements

Tone (professional/casual), Length (word count), and Exclusions.

The Golden Rule

Instructions on when to state limitations instead of guessing.

Application

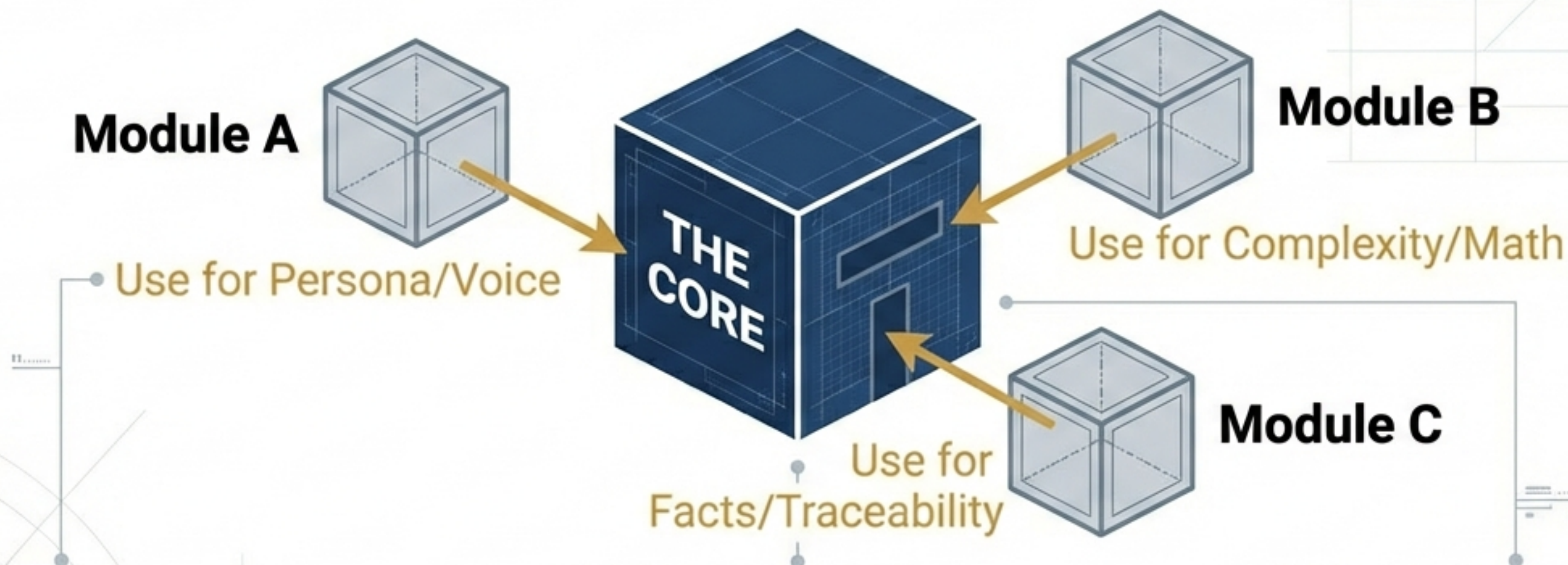
The Prompt Input

Constraints & Tone: Use a neutral, professional tone. Use at least 3 independent sources. Avoid opinion pieces and blogs. Do not fabricate citations. Keep the total response under 180 words.

 *Anti-Hallucination*

When the Core isn't enough.

Modules extend the Core without changing it.



Risk:
When mistakes are
costly (Medical, Legal).

Inconsistency:
When the model's
voice drifts.

Complexity:
When multi-step
reasoning is required.



Module A: Role & Persona

Theory

Function

Defines the AI's expertise and voice to improve adherence to the task.

How to Apply

Start with "You are a [Role]..." and add 1-2 specifics about domain and audience.

Application

Prompt Card

Optional Module Input

Engineering Gold
Specific expertise,
not just "Scientist"

Role: You are an **investigative fact-checker**. You use reputable databases, peer-reviewed studies, and official statistics to verify claims in Exoplanet atmospheres and the James Webb Space Telescope (JWST).



Module B: Reasoning & Quality Checks

Theory

Function

Adds a brief plan and a self-check step.
This reduces errors in multi-step work.

How to Apply

Request a short '**Verification Plan**' before the answer, and a '**Quality Check**' after.

Application

Optional Module Input

Prompt Card

Reasoning & Quality Checks:

Before writing the final answer, write a short "**Verification Plan**" describing how you will verify the claim.

After the **Sources** section, add a one-line "**Quality Check**" confirming:
(a) verdict matches evidence,
(b) word limits met,
(c) sources are independent.

The
Reliability
Loop

Start
End



Module C: Grounding & Sources

Theory

Function

Anchors factual claims in evidence.
Essential for traceability.

How to Apply

Specify allowed source types and add a fallback rule to state limitations.

Application

Optional Module Input

Prompt Card

Grounding & Sources: Prefer primary or authoritative sources. If credible evidence is insufficient, set Verdict to Unsubstantiated and state what evidence would be needed instead of guessing.

Fail-safe mechanism

The Machine in Motion: The Complete Prompt

Role

Role: You are an investigative fact-checker. You use reputable databases and primary sources to verify the accuracy of claims.

Context

Context: I will provide a claim to verify. Assume the audience is a typical knowledge worker. Avoid jargon where possible. Be concise.

Claim

Claim: "In 2022, NASA's James Webb Space Telescope detected carbon dioxide in the atmosphere of a planet outside our solar system for the first time."

Task (Goal)

Task (Goal): Determine whether the claim is True, False, or Unsubstantiated, and provide a justification based on evidence.

Output Format

Output Format: Use exactly these headings:

- 1) Verdict
- 2) Justification
- 3) Sources
- 4) Confidence Score (1-5)
- 5) Notes/Limitations. Do not add any other sections.

Modules

Reasoning: Add a short verification plan and a final self-check step.

Grounding: Prefer primary or authoritative sources. If evidence is insufficient, state limitations.

The Result: A Verifiable Truth.



Verification Plan:

- Confirm claim details in NASA and ESA Webb releases.
- Verify peer-reviewed evidence of CO₂ detection in WASP-39 b via JWST spectroscopy.
- Cross-check dates and instruments.



Verdict: True



Evidence Summary:

NASA and ESA announced that JWST observed the transiting exoplanet WASP-39 b and found clear spectroscopic evidence of carbon dioxide (CO₂) in its atmosphere.



Sources:

1. NASA. (2022). NASA's Webb detects carbon dioxide...
2. ESA. (2022). Webb detects carbon dioxide...
3. Nature. (2023). Identification of carbon dioxide...



Quality Check:

Verdict matches evidence; word limits met; sources are independent.

The Human Layer: Trust but Verify.

The structure allows for rapid human audit.



User Icon

Check Verification
Plan vs Summary

Verification Plan:

- Confirm details in NASA and ESA Web releases.
- Verify peer-reviewed evidence of CO₂ detection in WASP-39 via NIST spectroscopy.
- Cross-check dates and instruments.

Verdict: True

Evidence Summary:

NASA and ESA announced that JWST observed the transiting exoplanet WASP-39 b and found clear spectroscopic evidence of carbon dioxide (CO₂) in its atmosphere.

Sources:

1. NASA, (2022). NASA Announces Carbon Dioxide...
2. ESA, (2022). Robust detection of carbon dioxide...
3. Nature, (2022). New evidence of carbon dioxide...

Quality Check:

Best practices: evidence, word links and sources are independent.

Click Links
to Confirm



Source
Icon

Audit Complete

Takeaway: The user can now quickly cross-check the AI's summary against the provided high-quality sources, ensuring the response is grounded in verifiable information.

V2 Framework Cheat Sheet



The Core (Always Include)

1. **Context:** Background, Audience, Scope.
2. **Task:** Action Verbs, Deliverables.
3. **Output Format:** Structure, Headers, Schema.
4. **Constraints:** Tone, Limits, Safety.



The Modules (Use as Needed)

1. **Role:** Expertise & Voice.
2. **Reasoning:** Plans & Self-Checks.
3. **Grounding:** Sources & Evidence.

Golden Rule: If credible evidence is insufficient, the model must state the limitation rather than guess.